

# MODERN OPHTHALMIC OPTICS

JOSÉ ALONSO

*Indizen Optical Technologies*

JOSÉ A. GÓMEZ-PEDRERO

*Complutense University of Madrid*

JUAN A. QUIROGA

*Indizen Optical Technologies*

|     |                     |    |
|-----|---------------------|----|
| 1   | Introduction        | ix |
| 2   | Geometrical Optics  | 1  |
| 3   | Optical Systems     | 1  |
| 4   | Optical Instruments | 4  |
| 5   | Optical Systems     | 11 |
| 6   | Optical Systems     | 13 |
| 7   | Optical Systems     | 16 |
| 8   | Optical Systems     | 21 |
| 9   | Optical Systems     | 21 |
| 10  | Optical Systems     | 24 |
| 11  | Optical Systems     | 36 |
| 12  | Optical Systems     | 54 |
| 13  | Optical Systems     | 62 |
| 14  | Optical Systems     | 62 |
| 15  | Optical Systems     | 62 |
| 16  | Optical Systems     | 66 |
| 17  | Optical Systems     | 73 |
| 18  | Optical Systems     | 80 |
| 19  | Optical Systems     | 80 |
| 20  | Optical Systems     | 82 |
| 21  | Optical Systems     | 93 |
| 22  | Optical Systems     | 93 |
| 23  | Optical Systems     | 93 |
| 24  | Optical Systems     | 93 |
| 25  | Optical Systems     | 93 |
| 26  | Optical Systems     | 93 |
| 27  | Optical Systems     | 93 |
| 28  | Optical Systems     | 93 |
| 29  | Optical Systems     | 93 |
| 30  | Optical Systems     | 93 |
| 31  | Optical Systems     | 93 |
| 32  | Optical Systems     | 93 |
| 33  | Optical Systems     | 93 |
| 34  | Optical Systems     | 93 |
| 35  | Optical Systems     | 93 |
| 36  | Optical Systems     | 93 |
| 37  | Optical Systems     | 93 |
| 38  | Optical Systems     | 93 |
| 39  | Optical Systems     | 93 |
| 40  | Optical Systems     | 93 |
| 41  | Optical Systems     | 93 |
| 42  | Optical Systems     | 93 |
| 43  | Optical Systems     | 93 |
| 44  | Optical Systems     | 93 |
| 45  | Optical Systems     | 93 |
| 46  | Optical Systems     | 93 |
| 47  | Optical Systems     | 93 |
| 48  | Optical Systems     | 93 |
| 49  | Optical Systems     | 93 |
| 50  | Optical Systems     | 93 |
| 51  | Optical Systems     | 93 |
| 52  | Optical Systems     | 93 |
| 53  | Optical Systems     | 93 |
| 54  | Optical Systems     | 93 |
| 55  | Optical Systems     | 93 |
| 56  | Optical Systems     | 93 |
| 57  | Optical Systems     | 93 |
| 58  | Optical Systems     | 93 |
| 59  | Optical Systems     | 93 |
| 60  | Optical Systems     | 93 |
| 61  | Optical Systems     | 93 |
| 62  | Optical Systems     | 93 |
| 63  | Optical Systems     | 93 |
| 64  | Optical Systems     | 93 |
| 65  | Optical Systems     | 93 |
| 66  | Optical Systems     | 93 |
| 67  | Optical Systems     | 93 |
| 68  | Optical Systems     | 93 |
| 69  | Optical Systems     | 93 |
| 70  | Optical Systems     | 93 |
| 71  | Optical Systems     | 93 |
| 72  | Optical Systems     | 93 |
| 73  | Optical Systems     | 93 |
| 74  | Optical Systems     | 93 |
| 75  | Optical Systems     | 93 |
| 76  | Optical Systems     | 93 |
| 77  | Optical Systems     | 93 |
| 78  | Optical Systems     | 93 |
| 79  | Optical Systems     | 93 |
| 80  | Optical Systems     | 93 |
| 81  | Optical Systems     | 93 |
| 82  | Optical Systems     | 93 |
| 83  | Optical Systems     | 93 |
| 84  | Optical Systems     | 93 |
| 85  | Optical Systems     | 93 |
| 86  | Optical Systems     | 93 |
| 87  | Optical Systems     | 93 |
| 88  | Optical Systems     | 93 |
| 89  | Optical Systems     | 93 |
| 90  | Optical Systems     | 93 |
| 91  | Optical Systems     | 93 |
| 92  | Optical Systems     | 93 |
| 93  | Optical Systems     | 93 |
| 94  | Optical Systems     | 93 |
| 95  | Optical Systems     | 93 |
| 96  | Optical Systems     | 93 |
| 97  | Optical Systems     | 93 |
| 98  | Optical Systems     | 93 |
| 99  | Optical Systems     | 93 |
| 100 | Optical Systems     | 93 |



CAMBRIDGE  
UNIVERSITY PRESS

University Printing House, Cambridge CB2 8BS, United Kingdom

One Liberty Plaza, 20th Floor, New York, NY 10006, USA

477 Williamstown Road, Port Melbourne, VIC 3207, Australia

314–321, 3rd Floor, Plot 3, Splendor Forum, Jasola District Centre, New Delhi – 110025, India

79 Anson Road, #06–04/06, Singapore 079906

Cambridge University Press is part of the University of Cambridge.

It furthers the University's mission by disseminating knowledge in the pursuit of education, learning, and research at the highest international levels of excellence.

www.cambridge.org

Information on this title: www.cambridge.org/9781107110748

DOI: 10.1017/9781316275474

© José Alonso, José A. Gómez-Pedrero, and Juan A. Quiroga 2019

This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press.

First published 2019

Printed in the United Kingdom by TJ International Ltd, Padstow Cornwall

A catalogue record for this publication is available from the British Library.

Library of Congress Cataloging-in-Publication Data

Names: Alonso, José, 1965– author. | Gómez-Pedrero, José A., 1971– author. |

Quiroga, J. Antonio (Juan Antonio) author.

Title: Modern ophthalmic optics / José Alonso, Indizen Optical Technologies, Madrid,

José A. Gómez-Pedrero, Universidad Complutense, Madrid,

Juan A. Quiroga, Indizen Optical Technologies, Madrid.

Description: Cambridge, United Kingdom ; New York, NY :

Cambridge University Press, 2019. | Includes bibliographical references and index.

Identifiers: LCCN 2018045074 | ISBN 9781107110748 (hardback)

Subjects: LCSH: Ophthalmic lenses. | Ophthalmology. | Eyeglasses. | BISAC: SCIENCE / Optics.

Classification: LCC RE976 .A46 2019 | DDC 617.7/52–dc23

LC record available at <https://lccn.loc.gov/2018045074>

ISBN 978-1-107-11074-8 Hardback

Cambridge University Press has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication and does not guarantee that any content on such websites is, or will remain, accurate or appropriate.



## Contents

|                                                                         |         |
|-------------------------------------------------------------------------|---------|
| <i>Foreword by Dr. M. Morris</i>                                        | page ix |
| <i>Preface</i>                                                          | xi      |
| <i>Acknowledgments</i>                                                  | xv      |
| <i>List of Symbols and Acronyms</i>                                     | xvi     |
| <b>1 Ophthalmic Materials</b>                                           | 1       |
| 1.1 Introduction                                                        | 1       |
| 1.2 History of Glass and Glass Manufacturing                            | 1       |
| 1.3 Glass Properties                                                    | 4       |
| 1.4 History of the Optical Polymers Employed in the Ophthalmic Industry | 11      |
| 1.5 Properties of Optical Polymers                                      | 13      |
| 1.6 Summary of Ophthalmic Materials                                     | 16      |
| <b>2 Surfaces in Ophthalmic Lenses</b>                                  | 21      |
| 2.1 Introduction                                                        | 21      |
| 2.2 Surfaces with Revolution Symmetry                                   | 24      |
| 2.3 Surfaces with Axial Symmetry                                        | 36      |
| 2.4 Surfaces with Many Degrees of Freedom                               | 54      |
| <b>3 Wavefronts and Rays</b>                                            | 62      |
| 3.1 Introduction                                                        | 62      |
| 3.2 Vergence and Wavefront                                              | 62      |
| 3.3 Ray Propagation                                                     | 66      |
| 3.4 Refraction                                                          | 73      |
| <b>4 Single Vision Lenses</b>                                           | 80      |
| 4.1 Introduction                                                        | 80      |
| 4.2 Geometrical Aspects                                                 | 82      |
| 4.3 Paraxial Optical Properties                                         | 93      |
| <b>5 The Lens-Eye System</b>                                            | 123     |
| 5.1 Introduction                                                        | 123     |

|      |                                                 |     |
|------|-------------------------------------------------|-----|
| 5.2  | Basic Optical Physiology                        | 124 |
| 5.3  | Compensation of Refractive Errors               | 147 |
| 5.4  | Prismatic Effects                               | 157 |
| 5.5  | Magnification and Field of View                 | 189 |
| 6    | Aberrations and Lens Design                     | 205 |
| 6.1  | Introduction                                    | 205 |
| 6.2  | Aberrations of the Lens-Eye System              | 207 |
| 6.3  | Classical Theory of Ophthalmic Lens Design      | 212 |
| 6.4  | Modern ("Free-Form") Lens Design                | 231 |
| 7    | Optics of Contact and Intraocular Lenses        | 236 |
| 7.1  | Introduction                                    | 236 |
| 7.2  | Optical Properties of Contact Lenses            | 237 |
| 7.3  | Multifocal Contact Lens Designs                 | 245 |
| 7.4  | Optical Properties of Intraocular Lenses        | 247 |
| 7.5  | Design of Multifocal Intraocular Lenses         | 251 |
| 8    | Multifocal Lenses                               | 255 |
| 8.1  | Introduction                                    | 255 |
| 8.2  | Presbyopia and the Compensation of Near Vision  | 256 |
| 8.3  | Bifocal Lenses                                  | 265 |
| 8.4  | Progressive Lenses                              | 284 |
| 9    | Low Vision Aids and High Power Lenses           | 329 |
| 9.1  | Introduction                                    | 329 |
| 9.2  | The Problem of Low Vision Compensation          | 330 |
| 9.3  | Low Vision Aids for Close Objects: Magnifiers   | 336 |
| 9.4  | Low Vision Aids for Distant Objects: Telescopes | 343 |
| 9.5  | Low Vision Aids: Field Increasing Aids          | 351 |
| 9.6  | High Power Ophthalmic Lenses                    | 352 |
| 10   | Lens Manufacturing and Measurement              | 355 |
| 10.1 | Introduction                                    | 355 |
| 10.2 | Lens Surfacing                                  | 357 |
| 10.3 | Free-Form Lens Manufacturing                    | 366 |
| 10.4 | Lens Measurement                                | 370 |
| 11   | Filters and Coatings                            | 405 |
| 11.1 | Introduction                                    | 405 |
| 11.2 | Ocular Hazards Due to Electromagnetic Radiation | 405 |
| 11.3 | Filters for Ocular Protection                   | 410 |
| 11.4 | Anti-Reflective Coatings                        | 425 |
| 11.5 | Other Coatings                                  | 436 |

|            |                                            |     |
|------------|--------------------------------------------|-----|
| Appendix A | Frames                                     | 442 |
| Appendix B | Introduction to Matrix Algebra             | 460 |
| Appendix C | Introduction to Surface Geometry           | 468 |
| Appendix D | Local Dioptric Power Matrix                | 488 |
| Appendix E | Seidel Aberrations and Zernike Polynomials | 492 |
| Appendix F | Abelès Theory of Multilayer Films          | 511 |
| References |                                            | 520 |
| Index      |                                            | 542 |